

Questions with Answer Keys

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Q1: 24 Feb (Shift 1) - Single Correct

Which of the following ore is concentrated using group 1 cyanide salt?

- (1) Sphalerite
- (2) Siderite
- (3) Malachite
- (4) Calamine

Q2: 24 Feb (Shift 1) - Single Correct

The major components in "Gun Metal" are:

- (1) Al, Cu, Mg and Mn
- (2) Cu, Sn and Zn
- (3) Cu, Zn and Ni
- (4) Cu, Ni and Fe

Q3: 24 Feb (Shift 1) - Single Correct

Al_2O_3 was leached with alkali to get X. The solution of X on passing of gas Y, forms Z. X, Y and Z respectively are :

- (1) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (2) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{SO}_2$, $\text{Z} = \text{Al}_2\text{O}_3$
- (3) $\text{X} = \text{Al}(\text{OH})_3$, $\text{Y} = \text{SO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (4) $\text{X} = \text{Al}(\text{OH})_3$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3$

Q4: 24 Feb (Shift 2) - Single Correct

Questions with Answer Keys

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Match List-I with List-II

List- I

(Metal)

(a) Aluminium

(b) Iron

(c) Copper

(d) Zinc

List-II

(Ores)

(i) Siderite

(ii) Calamine

(iii) Kaolinite

(iv) Malachite

Choose the correct answer from the options given below:

(1) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(2) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

(3) (a) – (iii), (b) – (i), (c) – (iv), (d) – (ii)

(4) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)

Q5: 25 Feb (Shift 1) - Single Correct

Ellingham diagram is a graphical representation of:

(1) ΔG vs T(2) $(\Delta G - T\Delta S)$ vs T(3) ΔH vs T(4) ΔG vs P

Q6: 25 Feb (Shift 2) - Single Correct

The major components of German Silver are :

(1) Cu, Zn and Ag

(2) Ge, Cu and Ag

(3) Zn, Ni and Ag

(4) Cu, Zn and Ni

Questions with Answer Keys

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Q7: 25 Feb (Shift 2) - Single Correct

The method used for the purification of Indium is :

- (1) van Arkel method
- (2) vapour phase refining
- (3) zone refining
- (4) Liquation

Q8: 26 Feb (Shift 1) - Single Correct

Match List-I with List-II.

List -I**(Ore)****List-II****(Element Present)**

- | | |
|-----------------|----------------|
| (a) Kernite | (i) Tin |
| (b) Cassiterite | (ii) Boron |
| (c) Calamine | (iii) Fluorine |
| (d) Cryolite | (iv) Zinc |

Choose the most appropriate answer from the option given below :

- (1) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
- (2) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)
- (3) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)
- (4) (a) – (iii), (b) – (i), (c) – (ii), (d) – (iv)

Q9: 26 Feb (Shift 2) - Single Correct

Match List-I with List-II

List-I

- (a) Siderite
- (b) Calamine
- (c) Malachite
- (d) Cryolite

List-II

- (i) Cu
- (ii) Ca
- (iii) Fe
- (iv) Al
- (v) Zn

Choose the correct answer from the options given below:

(1) (a) – (i), (b) – (ii), (c) – (v), (d) – (iii)

(2) (a) – (iii), (b) – (v), (c) – (i), (d) – (iv)

(3) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

(4) (a) – (iii), (b) – (i), (c) – (v), (d) – (ii)

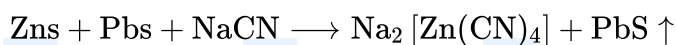
Answer Key**Q1 (1)****Q2 (2)****Q3 (1)****Q4 (3)****Q5 (1)****Q6 (4)****Q7 (3)****Q8 (2)****Q9 (2)**

Hints and Solutions

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Q1 (1)

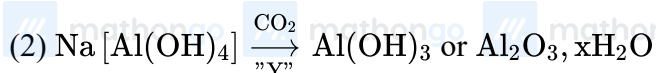
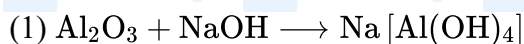
Conc. of sphalerite, first by cyanide salt as a depressant to remove the impurity of galena



Q2 (2)

Gun metal" is alloy of copper with tin and zinc.

Q3 (1)



Q4 (3)

Siderite FeCO_3

Calamine ZnCO_3

Kaolinite $\text{Si}_2\text{Al}_2\text{O}_5(\text{OH})_4$ or $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$

Malachite $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$

Q5 (1)

Ellingham diagram tells us about the spontaneity of a reaction with temperature.

Q6 (4)

Fact

German silver is alloy which does not have silver.

$\text{Cu} - 50\%$; $\text{Ni} - 30\%$; $\text{Zn} - 20\%$

Q7 (3)

Fact

Ga, In, Si, Ge are refined by zone refining or vaccume refining.

Q8 (2)

Hints and Solutions

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Fact

Q9 (2)

Si — FeCO_3

Calamine — ZnCO_3

Malachite — $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$

Cryolite — Na_3AlF_6